AIMI: Agentic In-Hand Manipulation for Language-Guided Visual Exploration

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Figure 1. **Language-guided visual exploration with AIMI.** The agent chooses how to inspect an object, uses new views to identify the requested face, and guides its presentation. Staged simulation views illustrate the task; recorded trials are reported separately.

Abstract

Dexterous in-hand manipulation is an important skill for robots to reposition and use objects. However, a target specified by language may initially be hidden or blocked by the fingers, requiring the robot to explore before presenting it. In this work, we present AIMI, an agentic in-hand manipulation framework for language-guided visual exploration. The agent chooses inspection actions, identifies the target from new images, and selects a motor goal. Local controllers adjust the motor policy's commands during reasoning and execution. We evaluate our framework in simulation on six objects, each with its own motor policy. Under changed target-to-skill mappings, it achieves 65.8% mean success, exceeding the strongest evaluated baseline by 40.4 percentage points. We demonstrate sim-to-real transfer on four real-world objects using RGB pose feedback, achieving 87.5% success over 80 trials, compared with 66.25% for the strongest evaluated baseline.

1. Introduction

Dexterous in-hand manipulation allows robots to adjust and use objects without releasing their grasp. It also offers a way to inspect surfaces that a camera cannot initially see. Consider the instruction "present the face marked T": the target may face away from the camera or be blocked by the fingers. The robot must move the object to find T, identify it from new observations, and present it while maintaining its grasp.

Recent methods learn dexterous skills conditioned on a desired pose or goal image [2, 3, 11]. Choosing such a goal from language is difficult when the requested surface is not visible. A robot must decide how to inspect the object and use what it sees to guide later actions. This creates a second challenge: semantic reasoning takes time, while the hand needs continuous feedback to control the object.

In this work, we present AIMI (**A**gentic **I**n-hand **M**anipulation for **I**nspection). Our key idea is to use hand motion both to gather evidence about the requested target and to act on that evidence. Given language, images and action history, an agent chooses inspection actions and reviews their outcomes. Once it identifies the target, it links that tar-

get to a motor goal, which we call *target binding*, and guides its presentation. This repeated observation–decision–action loop makes exploration part of the manipulation process.

We build on ViserDex's teacher–student foundation [3], adapted to Revo3. Teacher-supervised adaptation connects the student's motor skills to goal-bank vectors selected by the agent. Local controllers adjust the student's joint commands during reasoning and execution. While an agent reply is pending, learned speed adjustment and joint correction let the hand respond to feedback without waiting for a new decision.

Our contributions are as follows:

- We introduce an agentic framework for language-guided in-hand visual exploration, connecting inspection, target binding and presentation through successive observations.
- We develop a goal-conditioned control interface with teacher-supervised local controllers. Joint correction complements speed adjustment to support continuous manipulation during agent reasoning.
- We evaluate six object-specific simulation systems, achieving 65.8% mean success under changed target-to-skill mappings. Controlled studies show how fresh images improve target binding and joint correction improves completion. We further evaluate English and Chinese instructions on three objects and demonstrate sim-to-real transfer on four real-world objects with 87.5% success over 80 trials.

2. Related Work

Agentic reasoning and embodied feedback. SayCan selects skills using robotic affordances [10], while Inner Monologue updates plans from environmental feedback [8]. Code as Policies connects perception and control APIs [14]; VoxPoser and ReKep ground language in spatial value maps and keypoint constraints [7, 9]. Magma connects multimodal representations to actions [20], and VERA includes dexterous reorientation from visual plans [13]. We focus on acquiring visual evidence through hand motion: the agent inspects an object to identify a requested target while local control maintains the ongoing manipulation. Our image and inspection-policy interventions examine how these observations affect later decisions.

Visual and goal-conditioned dexterity. ViserDex combines monocular pose estimation with recurrent teacher–student control [3]. Our Revo3 motor foundation adapts this approach. GoalDex learns from current and goal images, with a VLM selecting visual goals from language [2]. Our framework addresses the preceding observation problem: the requested face may need to be exposed before a presentation goal can be chosen. It connects this exploration to goal-bank-conditioned control. ManipTrans uses residual learning for bimanual transfer [12]; our corrections support inspection and presentation within one hand.

RL for dexterous in-hand manipulation. Simulation-trained policies reorient objects on physical hands through domain randomization [1], handle diverse shapes [6], and adapt to changing object properties [15]. RotateIt combines vision, touch and proprioception through distillation [16], while AnyRotate supports different rotation axes and hand orientations [21]. SimToolReal trains a goal-conditioned policy on procedural objects for transfer to tool manipulation [11]. We build on these motor capabilities to connect language, visual exploration and target-directed manipulation.

Asynchronous robot policies. Prior work addresses inference delay through real-time action chunks [4], online corrections [18], latency-aware RL [22], and temporal counterfactual preferences [23]. LiMA combines sparse semantic decisions with asynchronous dexterous actions [5]. Our focus is their use during in-hand visual exploration: local controllers adjust a motor policy's commands while an agent gathers evidence and selects goals. The controller study separates scalar slowing from joint correction; the visual studies examine observation and inspection choices.

3. Task and Evaluation Protocol

Exploration and target presentation. Given an object held in the hand and a language instruction, the robot must inspect the object as needed, identify the requested surface target, and present it without dropping the object. Exploration serves this target-specific objective. Q1 evaluates a marked cube, a wooden die, a duck, a hedgehog, a pig, and a milk carton in simulation; Q2 and Q3 use the cube. Instructions request one of six surface letters, or one through six dots on the die. The cube letters are $\{T, E, R, L, S, H\}$. Each target has a fixed reference orientation $Q(g)$ that places the requested face upward. Simulation methods are scored against this full-quaternion reference. Q4 reports the physical-hand protocol separately.

At step t, the agent receives an RGB image I_t and instruction ℓ. The motor policy receives proprioception x_t and measured pose $\hat{p}_t$. In simulation, cube pose is estimated from RGB; other objects use noisy measurements constructed from simulator pose. Physical-hand experiments use RGB pose estimates for all four objects. Controllers do not directly receive the unperturbed true pose p_t. In simulation, teacher actions and privileged targets are used for training, and true pose is used for independent scoring. During deployment, the controller cannot use a future agent reply. We denote the active goal vector by g_t, the recurrent state by h_t, and a pending request by $m_t = 1$.

Simulation timing. Only one agent request can be pending at a time. Cube episodes allow up to six model calls and

3,600 control steps at 30 Hz. Other objects use 20 Hz over the same 120 seconds of simulated time. For the cube, a call taking L wall-clock seconds can affect control only after it completes and at least $\lceil 30L \rceil$ steps have passed since the request. Physics continues while the agent reasons. The simulator runs slower than real time, so this models inference delay but does not demonstrate a 30 Hz hardware system.

Simulation completion criterion. All cube systems use the same stopping check at 3 Hz. The agent must have bound the requested symbol to a goal, with no request pending and no inspection underway. Measured orientation error must be at most $\epsilon = 0.16$ radians. This check can stop the system without an explicit stop action from the agent. On other objects, AIMI uses calibrated completion correction, while the baselines use a measured-pose check. Their stopping rules are detailed in the supplement, and all are scored with the same true-pose tolerance. At stop time τ, success is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared target binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. The target must be correct at the stop time, not merely reached earlier. We do not require an additional hold after stopping. A wrong binding or an incorrect final pose counts as failure even if the object remains in the hand. A drop terminates the trial when the object is at least 0.24 metres from its nominal grasp position.

Simulation trials and service failures. Unless stated otherwise, each condition uses 40 trials: 20 per seed under two seeds. Q2 instead uses 40 trials per seed under three seeds. Compared methods share initial physical states and scored targets. Control and feedback experiments also share instructions. In the mapping experiment, the requested letter changes with the layout so that the physical target stays the same. For service-eligible comparisons, a service error or a model call taking more than 120 seconds excludes the matched trial from every method. We retain all raw outcomes without replacement. Each reported rate uses its study-specific denominator, with service failures distinguished from control failures.

4. Method

AIMI connects an agent's exploration decisions to a continuously running motor policy. Training begins with a ViserDex-based teacher–student foundation adapted to Revo3 [3]. A frozen teacher then supervises student adaptation to a shared goal bank, followed by training local controllers around the adapted student. At deployment, new observations guide inspection and target binding (Fig. 2). We describe the cube implementation below; other objects have their own students, noisy pose feedback and maintenance controllers.

4.1. Teacher-supervised motor adaptation

We start from a pretrained student that takes geometric goals. A learned projection replaces these goal inputs with $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains proprioception and history without the target pose, and $u_t \in \mathbb{R}^{21}$ contains proposed joint targets in normalized coordinates. We reuse compatible encoders, recurrent and actor weights, and normalization statistics from the released policy. The adapted student receives neither the target quaternion nor the relative-goal quaternion.

We train the goal projection and adapt the recurrent policy using actions from the frozen teacher in a DAgger-style procedure [17]. A training-only decoder reconstructs normalized privileged state targets with the goal quaternion removed. Its MSE has weight 0.2 alongside the action MSE; goal-projection warmup uses action supervision alone. The decoder and teacher are absent at deployment.

For deployed cube control, a frozen CNN takes a 120×120 RGB crop and predicts image coordinates and depth for the object center and eight keypoints. Camera calibration and rigid fitting recover position, XYZW orientation and a fitting error. These images include hand occlusion. The student receives this estimated pose. Physical deployment follows the same RGB-to-pose-to-control interface. Following ViserDex, Gaussian-splat rendering supplies training images for the pose estimator; the deployed estimator recovers pose from camera RGB. Object-specific interfaces are described in the supplement.

4.2. Language-guided visual exploration

The cube has a bank of six normalized goal vectors $B = \{g_1, \ldots, g_6\}$, computed from cached SigLIP2 text features [19]. Each vector represents a color skill and its reference object orientation.

Inspection, binding and presentation. Given the instruction, current image and action history, the agent returns an explicit action: inspect, reorient, verify, stop, or abstain. We validate each action before applying it. During inspection, its color argument selects a goal vector from B to expose another view without declaring that the target has been found. After identifying the requested face, the agent binds it to a motor goal for presentation. It selects inspection and reorientation durations of 10–90 steps and uses subsequent images to update its decisions. These durations are separate from

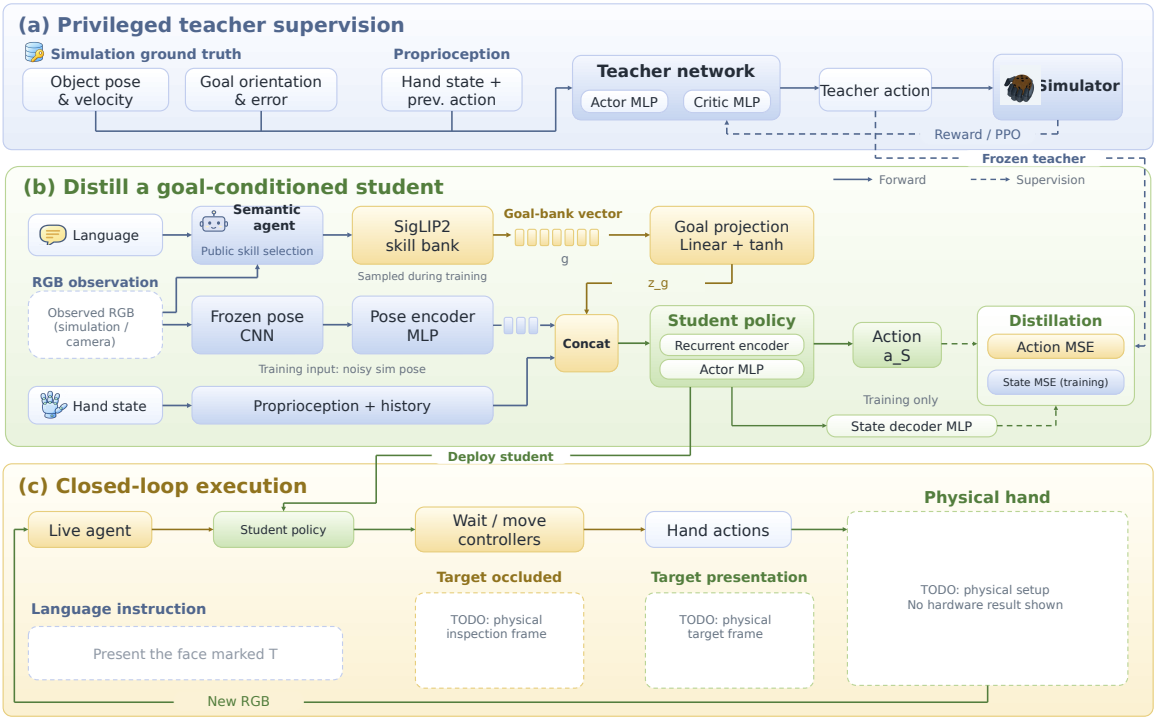


Figure 2. **AIMI training and deployment.** (a) Privileged teacher training follows the ViserDex-based Revo3 foundation. (b) The frozen teacher supervises student adaptation to goal-bank vectors; the state decoder is training-only. (c) An agent selects inspection or presentation skills from new observations while local controllers adjust joint commands. Dashed slots await physical-hand images.

model latency. The interface passes validated tool arguments through a fixed goal-bank lookup; it does not use the agent's hidden features.

4.3. Control while the agent is reasoning

The hand needs updated commands while an agent request is pending. The waiting controller receives a feature vector s_t containing proprioception, measured pose, the student's recurrent state and proposed action, fitting error, binding status and request status. It also receives the pose and command recorded when the request began, which we call the *request anchor*, and the elapsed waiting time. The active goal g_t is encoded separately. Before target binding, this goal represents the startup or inspection skill. The controller does not know the target that a future reply may select.

A fusion layer combines the outputs of a sensor encoder and a goal encoder. The waiting actor predicts a speed logit r_t and correction values $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ denote normalized current joint positions and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{4}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{5}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{6}$$

Zero actor outputs give fixed slowing with coefficient 0.15 while a request is pending. The environment keeps the same joint mapping and smoothing. Unlike scalar slowing, the joint correction can change the command direction as well as its size.

During training, the frozen teacher provides holding actions for the request anchor. At deployment, the actor uses only measured inputs and the active goal, without the pending reply. Writing the correction as $\delta_\phi = 0.35\tanh(v_t)$, we train it to match the teacher action while keeping this correction small:

$$\mathcal{L}_{\mathrm{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{7}$$

The waiting-actor update changes only control during pending requests. The student, pose estimator, execution controller and stopping rule remain fixed. Training details are in the supplement.

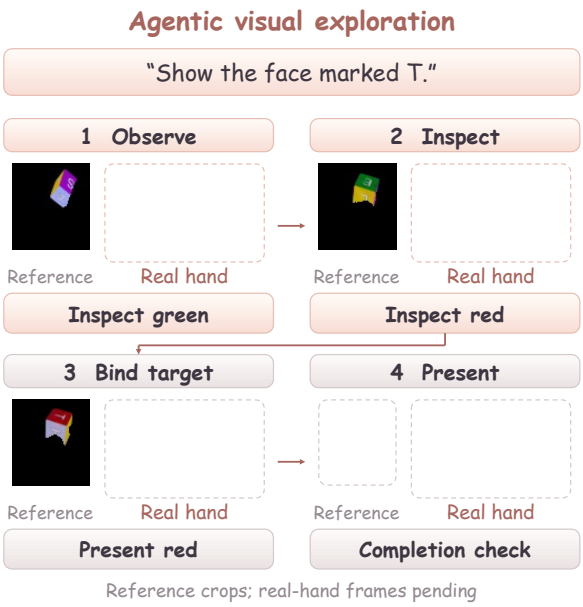


Figure 3. **Agentic visual exploration.** New observations guide inspection, target binding and presentation. Reference crops illustrate the sequence in simulation; physical-hand image slots remain open.

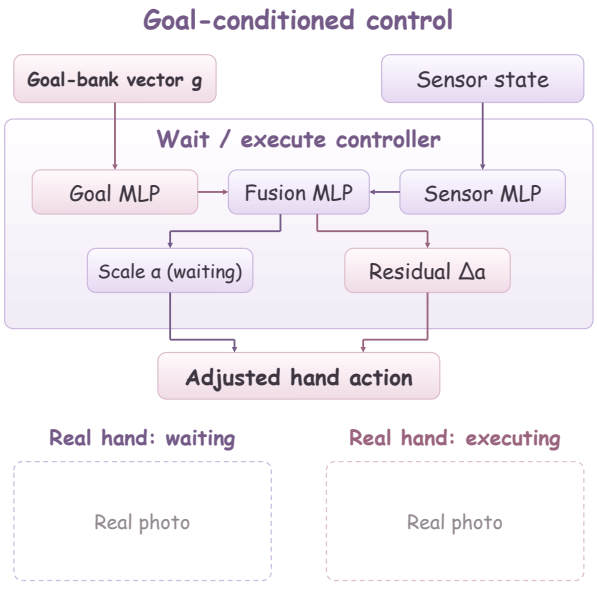


Figure 4. **Continuous local control.** Sensor features and the active goal-bank vector condition joint correction. The waiting controller also adjusts command speed. Empty slots are reserved for physical-hand images.

4.4. Control after an agent decision

When no request is pending ($m_t = 0$), a separate execution actor supplies v_t in Eq. 6. It learns from teacher actions to add a bounded correction to the student's proposal. The actor also receives the active goal, although our experiments do not isolate the benefit of this extra goal input.

4.5. Closed-loop execution

At each control step, the system updates the pose estimate, runs the recurrent student, selects the waiting or execution actor, and sends joint targets to the hand. Only a valid agent reply can change the active skill or target binding. Later images reflect the motion that occurred during reasoning and execution, closing the exploration loop. The measured-pose check in Sec. 3 determines completion. Invalid replies never trigger an oracle target selection.

We evaluate student adaptation separately from controller learning because a new student also changes the recurrent features and action proposals seen by the controllers.

5. Experiments

We organize the evidence around visual exploration (Q3), continuous control during reasoning (Q2), and complete-system performance across objects and instructions (Q1). Q2 and Q3 use the marked cube. Q4 evaluates perception for transfer and complete-system performance on four physical objects.

5.1. Setup and reporting

Simulation uses Revo3 assets in Isaac Lab 3.0/Newton. Unless specified otherwise, each simulated condition has 40 trials across two evaluation seeds, paired by physical task. Cube systems share the frozen RGB pose estimator, goal bank, student, stopping rule and 0.16-radian tolerance. Other simulated objects use separate students and noisy simulator pose. AIMI additionally uses calibrated completion correction on these objects. The VLM baselines omit the extra control actors and follow a fixed inspection schedule when the target is unreadable. AIMI chooses its inspection actions. These differences and measured inference delays are part of the complete-system comparison.

Q1 and Q3 compare methods on common service-eligible trials. Q2 reports all 120 trials per method and varies only control during pending requests. Its gains are measured within that cohort; Q1 uses different trials for complete-system comparison. The supplement gives study-specific seeds, paired tests and corrections for multiple comparisons.

5.2. Q3: Visual Exploration and Feedback

Three paired experiments examine how the agent acquires and uses visual evidence (Table 1): continuing to observe and decide, receiving new images, and choosing inspection direction and duration. Each comparison has its own trial set.

Condition	Success	%	Drops	Calls
(a) Continued feedback: one vs. up to six calls				
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120
(b) Fresh visual evidence: same six-call cap				
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98
(c) Inspection policy: same model and controllers				
Fixed sweep	31/40	77.5	1	126
Adaptive inspection	38/40	95.0	0	117

Table 1. **Q3: visual exploration on the cube.** Each panel schedules 40 trials per condition. (a) excludes one service-affected pair and reuses Q2's initial full-feedback subset. (b) changes only the agent's RGB input. (c) changes inspection direction and duration. (b,c) have no exclusions. Drops/calls include all trials; realized inference costs differ.

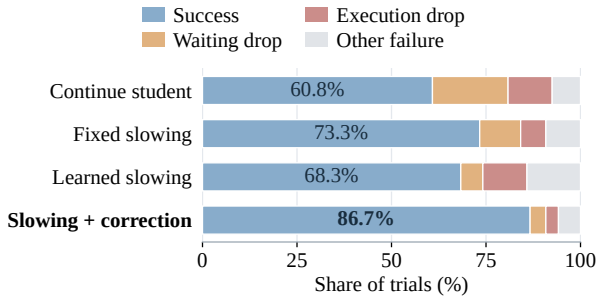


Figure 5. **Q2: outcomes during controller ablation.** Each bar contains 120 trials, 40 per evaluation seed. Labels show success rates; colored segments partition all outcomes. Drop stages record when failure occurs. Full counts are in the supplement.

(a) Does continued feedback help find the target? An initial plan selects up to six bounded actions from one image and measured feedback; continued feedback allows up to six calls with new images and history. Local controllers, stopping and time limits remain fixed. Success rises from 6/39 to 33/39 eligible trials, with all 27 discordant pairs favoring feedback (exact $p = 1.49 \times 10^{-8}$). Feedback resolves 22 of 28 initial-plan abstentions; 32 trials first bind after inspection (Fig. 6). The next comparison holds the call budget fixed.

(b) Does the next image provide useful evidence? Both conditions use the same six-call cap, request rule, live numeric feedback, history and local control. Only RGB changes: the current image versus the exact first frame. Success is 32/40 versus 9/40, with all 23 discordant pairs favoring fresh images (exact $p = 2.38 \times 10^{-7}$). Correct terminal bindings number 34 versus nine. Both first bind correctly on call one in the same nine trials; fresh images enable 25 additional correct bindings, 23 leading to success, versus none with repeated frames. Drops differ by one; initial visibility is not independently annotated.

(c) Does choosing the next view help? Forty new pairs compare agent-selected directions and durations with a fixed 90-step color sweep. Both use the same model, controllers, fresh images, history format and stopping; history records executed actions. The agent still decides whether to inspect, bind or abstain. Adaptive inspection achieves 38/40 successes (95.0%) versus 31/40 (77.5%). Nine pairs favor adaptive and two favor fixed inspection: a 17.5-point observed gain (exact $p = 0.0654$). Correct terminal bindings are 39 versus 32, with zero versus one drop. This trend remains statistically unresolved and covers one fixed sweep.

5.3. Q2: Control During Agent Reasoning

Figure 5 compares four ways to control the hand while the agent is reasoning: continue the student's command, slow it by a fixed coefficient of 0.15, learn the slowing coefficient, or learn slowing together with a bounded joint correction. All four use the same execution controller when no request is pending. Each method has 120 trials: 40 under each of three evaluation seeds, with the trained weights fixed. These seeds vary evaluation tasks, not training initialization.

Learned slowing with joint correction achieves 104/120 successes (86.7%), compared with 88/120 (73.3%) for fixed slowing, a gain of 13.3 percentage points. Waiting drops decrease from 13 to five, and execution drops from eight to four. Learning speed alone gives 82/120 successes (68.3%), below fixed slowing. The results favor learning speed and joint correction together. Drop stages tell us when failure occurs, but do not identify its cause. Seed settings and outcome breakdowns are in the supplement.

5.4. Q1: Objects, Target Mappings and Language

We compare original and changed target-to-skill mappings on six objects (Fig. 7), then test language and wording coverage. Separate native-image cohorts and their inconclusive AIMI–fine-tuned comparisons are reported in the supplement.

Aggregate system performance. Across 1,440 scheduled trials, mean success is 70.4% with original mappings and 65.8% with changed mappings, exceeding the strongest baseline means (44.2% and 25.4%) by 26.2 and 40.4 points. These descriptive, post-hoc means weight each object's service-eligible rate equally.

Four additional objects. Duck, hedgehog, pig and milk carton contribute 960 trials, with AIMI leading in each condition. Three service failures exclude matching trials from all

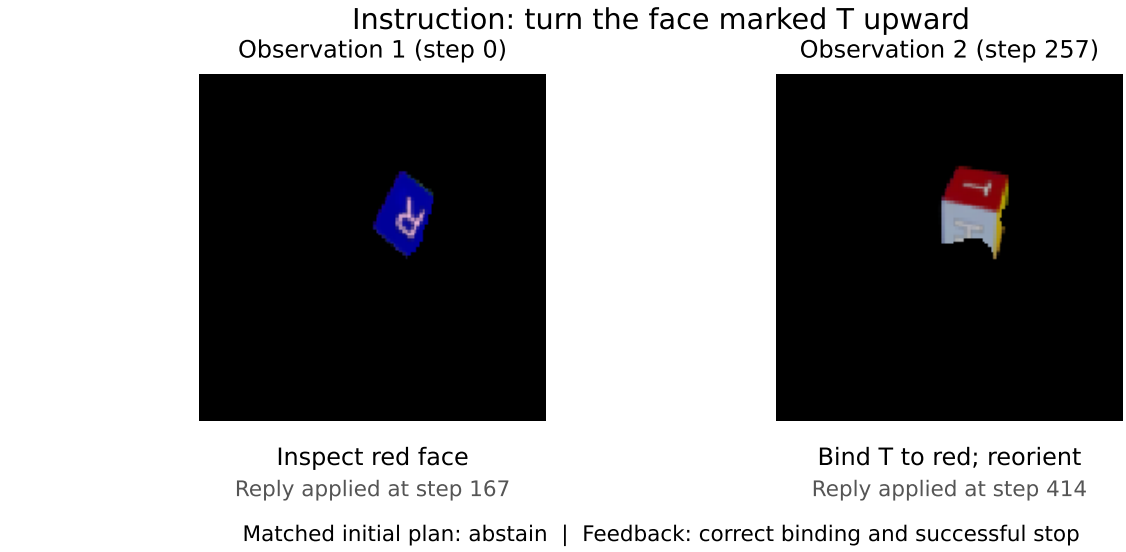


Figure 6. **New evidence enables target binding.** Unretouched model-input frames from a simulated trial. The initial plan abstains; continued feedback exposes T, after which AIMI binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from the same state and image. This selected example illustrates inspection before binding.

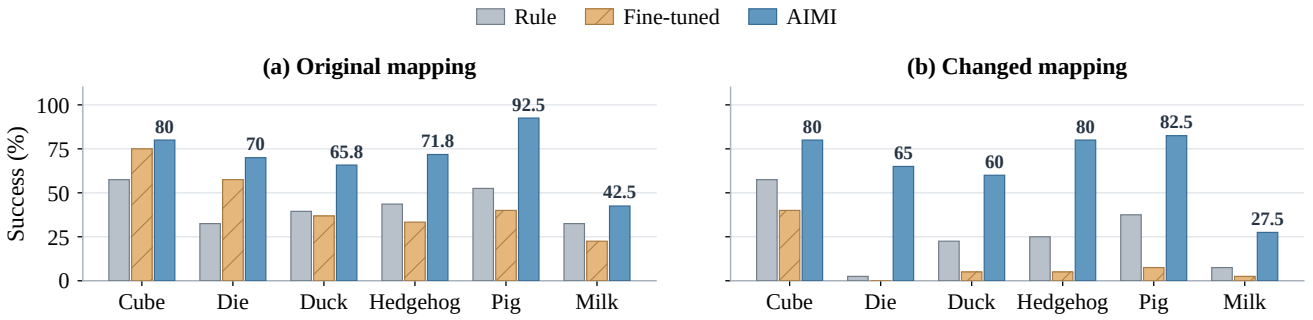


Figure 7. **Q1: task success across six simulated objects.** Bars use common service-eligible trials: 40 per cell, except 38 for original-mapping duck and 39 for original-mapping hedgehog. Labels show AIMI rates. Each object uses its own policy. The supplement retains all values and system differences. These are complete-system comparisons.

methods. Milk-carton drops remain frequent: 15/40 original-mapping and 20/40 changed-mapping AIMI trials. Reset checks and sensitivity analyses appear in the supplement.

Visual-semantic intervention. Only the agent's RGB changes across 40 matched states with original, reassigned or hidden letters; pose-estimator input and lighting remain fixed. Reassignment leaves AIMI at 32/40 successes but reduces fine-tuning from 30/40 to 16/40 (Holm $p = 0.0130$, six comparisons); rules remain at 23/40. Hiding letters gives 0/39, 3/40 and 0/40, respectively. Nine of eleven fine-tuned wrong bindings follow the old mapping. One reassignment does not establish broad generalization.

Dice layout transfer. Across 240 die trials, rotating the visual mesh preserves collision geometry, inertia and the skill-ID prompt. Exploratory paired tests favor AIMI over both baselines under the changed layout; its own layout difference remains inconclusive. The supplement reports all seven unadjusted contrasts.

Language and wording coverage. We test standard and rewritten English and Chinese instructions on three objects: 60 trials per condition across three seeds, totaling 2,160 trials (Table 2). Weights, mappings and scoring remain fixed, without Chinese retraining. The rule baseline also translates its task/output instructions. Raw outcomes remain in the supplement.

System	English		Chinese	
	Standard	Rewrite	Standard	Rewrite
Cube (n)	56	59	57	53
Rule VLM	48.2	45.8	31.6	32.1
Fine-tuned VLM	71.4	72.9	75.4	**81.1**
AIMI	**75.0**	**79.7**	**84.2**	67.9
Duck (n)	60	59	60	60
Rule VLM	50.0	25.4	33.3	31.7
Fine-tuned VLM	38.3	30.5	31.7	35.0
AIMI	**68.3**	**67.8**	**68.3**	**76.7**
Hedgehog (n)	60	60	52	60
Rule VLM	46.7	38.3	36.5	36.7
Fine-tuned VLM	38.3	31.7	34.6	25.0
AIMI	**73.3**	**76.7**	**75.0**	**76.7**

Table 2. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed n of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

Estimator	Pose recall (%) ↑		Invalid ↓
	10 mm / 10°	5 mm / 5°	
RGB	37.53	14.63	69
RGB-D	**64.73**	**47.56**	**1**

Table 3. **Q4: independent pose estimation.** Recall requires both errors below the listed thresholds. All 6,144 simulated observations remain in the denominator, including invalid predictions. Evaluation uses ideal masks and virtual depth.

AIMI exceeds fine-tuning in all eight duck/hedgehog conditions and rules in six after Holm correction over 36 method tests. Cube comparisons remain inconclusive, with fine-tuning leading for rewritten Chinese. Strict state/image pairing preserves these decisions. Language/wording contrasts remain inconclusive after a separate 36-test correction.

5.5. Q4: Pose Estimation and Sim-to-Real Transfer

Perception adaptation for transfer. We adapt the ViserDex-based perception pipeline with real-camera calibration and Gaussian-splat appearance assets. Table 3 evaluates RGB and RGB-D variants, showing improved pose recall with RGB-D in simulation. Training and camera adaptation differ between variants. This supporting perception study is separate from the RGB-pose hardware evaluation below.

Complete-system real-hand evaluation. We evaluate Rule, Fine-tuned and AIMI on a cube, die, ball and duck, with 20 trials per object and method (240 total). All use RGB-based pose estimation. A trial succeeds when the hand reaches the requested target and stops within 15° of its target orientation without dropping the object. Figure 8 shows per-object success. Overall rates are 40.0%, 66.25% and 87.5%, respectively. AIMI completes 70/80 trials, exceeding fine-tuning by 21.25 percentage points. Counts are provided in the supplement.

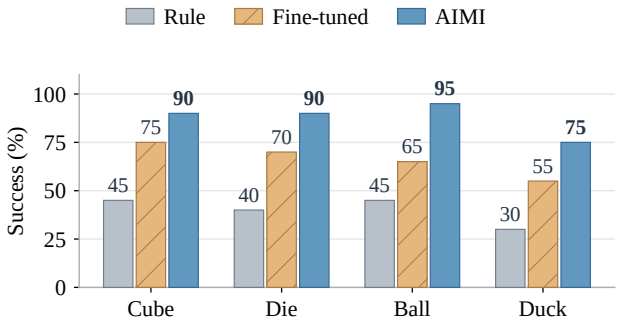


Figure 8. **Q4: complete-system success on the real hand.** Each bar contains 20 trials, using RGB pose feedback. Success requires reaching the target, stopping within 15°, and no drop.

6. Discussion and Limitations

Exploration and control. Q3 links new visual evidence to target binding and completion; the adaptive-inspection trend remains statistically unresolved. Q2 supports joint correction during reasoning. Q1 compares complete systems differing in models, history, controllers and stopping.

Evaluation limits. Equal call limits do not imply equal inference cost. Controller-goal conditioning remains unisolated, and Q3 tests one fixed sweep on one cube. Exploration is directed toward six predefined surface targets per object. Simulation completion measures binding and terminal pose; independent visibility and post-stop holding require separate reporting.

Deployment scope. Each object uses separate weights. Simulation uses RGB pose for cube and noisy simulator pose otherwise; hardware uses RGB pose. Hardware results do not establish unseen-object transfer or isolate perception gains. Real-camera pose accuracy remains unmeasured. Hosted-model changes limit reproducibility; service-eligible rates do not measure end-to-end availability.

7. Conclusion

AIMI connects language-guided visual exploration to continuous in-hand manipulation. New observations guide target identification while local controllers adjust motion. Cube studies support fresh visual feedback and joint correction during reasoning. Evaluations cover six simulated objects, two languages, and four physical objects with 87.5% success over 80 hardware trials.

LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. A hosted vision-language model and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

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